ELEKTRONIKA ANALOG

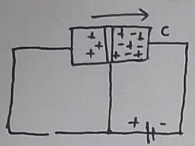
BAB 3

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2. what is the major difference between a bipolar and a unipolar device ?
   ↳ • transistor bipolar memanfaatkan elektron dan lubang dalam proses injeksi atau muatan yang terdiri dari 2 jenis transistor serta perubahan listrik dalam jumlah kecil.
   • transistor unbipolar hanya memanfaatkan salah satunya dari elektron dalam proses muatan

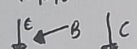
4. what is the source of the leakage current in a transistor ?
   ↳ sumber arus bocor (I_o) pada transistor ialah arus pembawa minoritas di kolektor

6. sketch a figure similiar to fig 3.4 for the reverse bias junction of an npn transistor.
   ↳



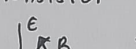
8. which of the transistor currents is always the largest ? which is always the smallest ? which two currents are relatively close in magnitude ?
   ↳ I_E terbesar $I_C \cong I_E$
   I_B terkecil

10. sketch the transistor symbol for a pnp and an npn transistor, and the insert the conventional flow direction for each current
    ↳ NPN



E = mengarah keluar
B = mengalir masuk ke basis
C = " " " kolektor

Transistor PNP



E = mengalir masuk
B = mengalir keluar basis
C = keluar dari kolektor

12. a). determine the average ac resistance for the characteristics of fig 3.10b
    ↳ $r_{av} = \frac{\Delta V}{\Delta I} = \frac{0,9V - 0,7V}{8mA - 0} = 25\,\Omega$

    b). for networks in which the magnitude of the resistive elements in typically in kilohms, in the approximation of fig 3.10c a valid one
    ↳ ya, karena nilai resistansi dari 25 Ω diabaikan dibanding dengan nilai resistansi lainnya

14. a). determine I_C if $V_{CB} = 10\,V$ and $V_{BE} = 800mV$
    ↳ using fig 3.7 first, $I_E \cong 7mA$
    then fig 3.8 result in $I_C \cong 7mA$

    b). determine V_{BE} if $I_C = 5mA$ and $V_{CB} = 10\,V$
    ↳ using fig 3.8 first, $I_E \cong 5mA$
    then fig 3.7 result in $V_{BE} \cong 0,78\,V$

    c). repeat part B using the characteristics of fig 3.10b
    ↳ $I_E = 5mA$ result in $V_{BE} \cong 0,81\,V$

d). Repeat part b

↳ $I_E = 5mA$ result in $V_{BE} \cong 0.7V$

e). compare the solutions for V_{BE}. can the diff be ignored if vatage levels greater than a few volts are typically encountered

↳ yes, the diff in levels V_{BE} can be ignored in the network

16. sketch the common base BJT transistor configuration (npn & pnp) and indicate the polarity of the applied biar and resulting current directions

↳ NPN

V-CB (+)

[C]

[B]

[E] → I-E

GND V-B

V-EB diberi agar Emitter lebih positif dari base V-CB diberi agar colector lebih positif dari base

PNP

V-CB

[C]

[B]

[E] ← I-E

GND V-EB

18. calculate the voltage gain ($Av = \frac{V_L}{V_I}$) for the network of Fig 3.12 if the source has an internal resistance of 100Ω in series with Vi

↳ $I_1 = \frac{V_i}{R_1 + R_2} = \frac{200mV}{20\Omega + 100\Omega} = 1.67mA$

$I_2 = I_1 = 1.67\ mA$

$V_2 = I \cdot R = (1.67MA)(5k\Omega) = 35V$

$AV = \frac{V_o}{I_o} = \frac{0.35V}{0.2V} = 41.75$

20. using the characteristics of Fig 3.14

a). Find the value of Ic corresponding to $V_{BE} + 750mV$ and $V_{CE} = +5V$

↳ $I_B = 35\mu A$ $I_C = 5.2\ mA$

b). Find the value V_{CE} and V_{BE} corresponding to $I_C = 3mA$ and $I_B = 30\ \mu A$

↳ $V_{CE} = 13V$ $V_{BE} = 0.75V$

22. a). using the characteristics of Fig 3.14a determine I_{CEO} at $V_{CE} = 10V$

↳ $I_{CEO} = B \cdot I_{CBO}$

$I_B = 0$

$I_{CBO} \cong 10\mu A$ untuk V_{CE} yg besar

b). determine dc at I_B 10 A and V_{CE} 10V

↳ $B = \frac{I_C}{I_B}$ $B_{CD} = \frac{1mA}{10\mu A} = 100$

c). using the determine in part b, calculate I_{CBO}

↳ $I_{CEO} \cong B \cdot I_{CBO}$

$I_{CBO} = \frac{I_{CEO}}{B_{AC}} = \frac{10\mu A}{100} = 0.1\ \mu A$

a). using the characteristics of Fig 3.14a, determine Bac at $I_B = 80\ \mu A$ $V_{CE} = 5V$

↳ $B_{AC} = \frac{\Delta I_C}{\Delta I_B} = \frac{7.5mA - 6.2mA}{90\mu A - 70\mu A} = 65$

b). repeat part a at $I_B = 5\ \mu A$ and $V_{CE} = 15V$

↳ $B_{AC} = \frac{\Delta I_C}{\Delta I_B} = \frac{1.2mA - 0.4mA}{10\mu A - 0\mu A} = 80$

c). repeat part a at $I_B = 30\mu A$ and $V_{CE} = 10V$

↳ $\beta_{ac} = \frac{\Delta I_C}{\Delta I_B} = \frac{4.2mA - 2.3mA}{40\mu A - 20\mu A} = 95$

d). Analisis perubahan β_{AC} pada grafik

↳ Nilai β_{ac} berubah dari titik ke titik, Nilai β_{ac} lebih tinggi pada arus baris yg kecil

e). membandingkan β_{ac} dan β_{dc} dari prob 23

↳ β_{dc} lebih stabil dibandingkan β_{ac}

26. a) Given that $\alpha_{dc} = 0,987$, determine the corresponding value of β_{dc}

↳ $\beta = \frac{\alpha}{1-\alpha} = \frac{0,987}{1-0,987} = 75,92$

b). $\beta_{dc} = 120$, determine corresponding

↳ $\alpha = \frac{\beta}{\beta+1} = \frac{120}{120+1} = 0,992$

c). $\beta_{dc} = 180$ and $I_C = 2,0mA$ Find I I_E and I_B

↳ $I_B = \frac{I_C}{\beta} = \frac{2mA}{180} = 11.11\mu A$

$I_E = I_C + I_B = 2mA + 11.11\mu A = 2.011\ mA$

28. 2An input voltage of 2 V rms is applied to the circuit of Fig 3.21. Assuming that the emitte voltage follows the base voltage exactly and that V_{be} (rms) $= 0,1V$, calculate the circuit voltage amplification ($A_V = V_o/V_i$) and emitter current for $R_E = 1k\Omega$

↳ $V_e = V_i - V_{be} = 2V - 0,1V = 1,9V$ $\quad I_e = \frac{V_e}{R_e} = \frac{1,9V}{1k\Omega} = 1,9mA$

$A_V = \frac{V_o}{V_i} = \frac{1,9V}{2V} = 0,95 = 1$

30. determine the region of operation for a transistor having the characteristics of Fig 3.10 if ~~$I_{Cmax} = 40mW = V_{CE}I_C$~~ $I_{Cmax} = 7mA$, $V_{CEmax} = 17V$, $P_{Cmax} = 40mW$

↳ $I_C = I_{Cmax}$, $V_{CE} = \frac{P_{cmax}}{I_{cmax}} = \frac{40mW}{7mA} = 5,71V$

$V_{CE} = V_{CEmax}$, $I_C = \frac{P_{Cmax}}{V_{CEmax}} = \frac{40mV}{17V} = 2,35mA$

$V_{CE} = V$, $I_C = \frac{P_{cmax}}{V_{CE}} = \frac{40mW}{V} = mA$

32). Referring to Fig 3.23 determine the temperature range for the device in degress fahrenheit temperature akan beroperasi dalam $-55°C \leq T_J \leq 150°C$

↳ $F = \frac{9}{5}°C + 32°$

$= \frac{9}{9}(150°C) + 32° = -67°F$

$°F = \frac{9}{5}(150°C) + 32° = 302°F$

→ jadi $-67°C \leq T_j \leq 302°C$

34). Based on the data of Fig 3.23 what is the expected value of I_{CEO} using the average value of B_{dc}

↳ $I_{CBO} = 50\mu A$ max

$B_{avg} = \frac{B_{min} + B_{max}}{2} = \frac{50 + 150}{2} = \frac{200}{2} = 100$

Jadi, ~~I_{CBO}~~ $I_{CEO} = B_{ICBO} = (100)(50nA) = 5\mu A$

36). determine ~~with~~ whether the input capacitance in the common base configuration increases or decreases with increasing levels of reserve bias potential. can you explain why ?

↳ ketika potensi bias balik meningkat besarnya, kapasitansi input cibo berkurang. menyebabkan lebar daerah penipisan menjadi meningkat sehingga mengurangi kapasitansi ($C = \varepsilon \frac{A}{d}$)

38). using the characteristics of Fig 3.23 determine the level of B_{dc} at $I_C = 10mA$ at the three levels of temperature appearing in the figure is the change significan for the spesificited temperature range ? Is it an element to be concernad about in the dengn proses

↳ At $I_C = 1mA$ $h_{FE} \cong 120$

at $I_C = 10mA$ $h_{FE} \cong 160$

grafik no 6

